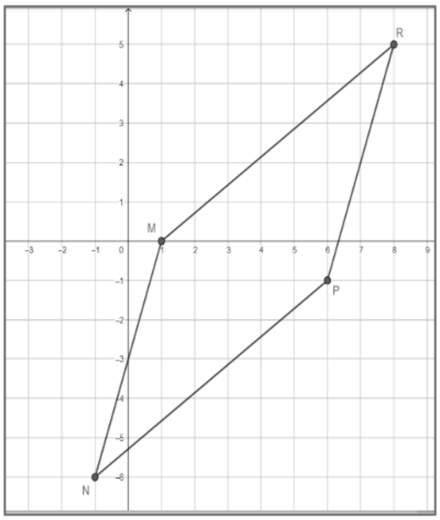
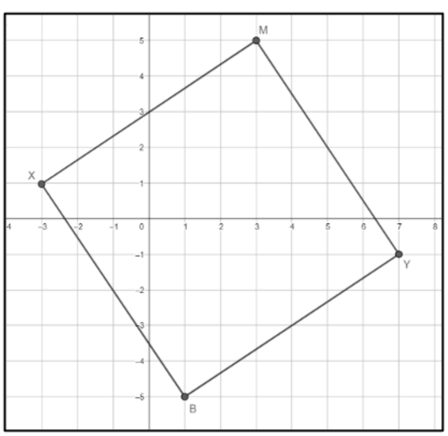
Geometry
Unit 13
Lesson 10 Practice

Name___Answer Key______
Date____________________
Period___________________

Complete the table for each image in questions 1 and 2.



Question 1	Work		Answer
	Slope	$\overline{MN}$ & $\overline{RP}$: $m = \frac{6}{2} = 3$ $\overline{MR}$ & $\overline{NP}$: $m = \frac{5}{7}$	**Properties** Opposite sides are parallel and congruent.
	Side Lengths	$MN = \sqrt{6^2 + 2^2} = 2\sqrt{10}$ $RP = \sqrt{2^2 + 6^2} = 2\sqrt{10}$ $MR = \sqrt{5^2 + 7^2} = \sqrt{74}$ $NP = \sqrt{7^2 + 5^2} = \sqrt{74}$	
	Diagonal Lengths	$NR = \sqrt{9^2 + 11^2} = \sqrt{202}$ $MP = \sqrt{1^2 + 5^2} = \sqrt{26}$	**Quadrilateral** Parallelogram



Question 2	Work		Answer
	Slope	$\overline{XM}$ & $\overline{BY}$: $m = \frac{4}{6} = \frac{2}{3}$ $\overline{XB}$ & $\overline{MY}$: $m = -\frac{6}{4} = -\frac{3}{2}$	**Properties** Opposite sides are parallel and congruent. Consecutive sides are perpendicular (90°). Diagonals are congruent. All sides are equal.
	Side Lengths	$\overline{XM}$, $\overline{MY}$, $\overline{YB}$, $\overline{BX}$: $= \sqrt{4^2 + 6^2} = \sqrt{52} = 2\sqrt{13}$	
	Diagonal Lengths	$MB = \sqrt{2^2 + 10^2} = 2\sqrt{26}$ $XY = \sqrt{10^2 + 2^2} = 2\sqrt{26}$	**Quadrilateral** Square

For questions 3 and 4, given the coordinates, classify the quadrilaterals based on their properties.

Question 3	**Work**		**Answer** **Properties**
$CMHG$ has vertices at $C(9,-1)$, $M(2,5)$, $H(-7,7)$, and $G(0,1)$. $CMHG$ is classified as a(n) rhombus because... all sides are equal, consecutive sides are not perpendicular.	**Slope**	$\overline{CM}$ & $\overline{HG}$: $m=-\frac{6}{7}$ $\overline{MH}$ & $\overline{CG}$: $m=-\frac{2}{9}$	Opposite sides are parallel and congruent. All sides are congruent. Diagonals are not congruent.
	Side Lengths	$CM=\sqrt{(-7)^2+6^2}=\sqrt{85}$ $MH=\sqrt{(-9)^2+2^2}=\sqrt{85}$ $HG=\sqrt{(-7)^2+6^2}=\sqrt{85}$ $GC=\sqrt{9^2+(-2)^2}=\sqrt{85}$	
	Diagonal Lengths	$CH=\sqrt{(-16)^2+8^2}=8\sqrt{5}$ $MG=\sqrt{2^2+4^2}=2\sqrt{5}$	**Quadrilateral** Rhombus

Question 4	**Work**		**Answer** **Properties**
$TCBY$ has vertices at $T(-3,-7)$, $C(5,-3)$, $B(3,1)$, and $Y(-5,-3)$. $TCBY$ is classified as a(n) rectangle because... opposite sides are parallel and congruent and all angles are right angles.	**Slope**	$\overline{TC}$ & $\overline{BY}$: $m=\frac{1}{2}$ $\overline{CB}$ & $\overline{YT}$: $m=-2$	Opposite sides are parallel and congruent. Consecutive sides are perpendicular. Diagonals are congruent.
	Side Lengths	$TC=\sqrt{8^2+4^2}=4\sqrt{5}$ $CB=\sqrt{(-2)^2+4^2}=2\sqrt{5}$ $BY=\sqrt{(-8)^2+(-4)^2}=4\sqrt{5}$ $YT=\sqrt{(-2)^2+4^2}=2\sqrt{5}$	
	Diagonal Lengths	$TB=\sqrt{6^2+8^2}=10$ $CY=\sqrt{10^2+0^2}=10$	**Quadrilateral** Rectangle